

Case Study 13

Brain Damage and Ethical Behaviour

Do we have a choice regarding how to behave or are we products of our genetic coding? The nature-nurture debate is classic in the field of psychology. Many people believe behaviour results from a blend of genetic coding (nature), and familial upbringing (how we were raised as children) -- nurture. A study published in the journal *Nature* suggests that ethical behaviour may be, at least in part, hardwired in the brain.

The Dilemma

American researchers presented a controversial scenario to 30 test subjects. The participants were told that if they do harm to one person, they could prevent future harm to many others. For example, if you knew that someone with a deadly infectious disease was planning to infect others (some of whom would die), would you kill the person or would you let the person intentionally infect others?

The Results



Many of the participants struggled in making a decision and most said they would not kill the infected individual. Six participants, however, stated a willingness to harm one individual to protect others. These six participants had damage to a part of their brains called the ventromedial prefrontal cortex (VMPC). The VMPC is part of the frontal lobe and the frontal lobe is linked to higher cognitive functioning.

Ralph Adolphs, professor of psychology and neuroscience at the California Institute of Technology, believes that VMPC brain damage results in abnormal social emotions. The six participants, he believed, lacked empathy and compassion. Of the remaining 24 participants, 12 had no brain damage and 12 had brain damage in areas other than the VMPC.

Although correlation is not causation, this study implies that aversion to hurting others may be hard-wired in the frontal lobe of the brain. This leaves researchers with an important question: Are social emotions necessary for moral judgments? The research continues!